$\mathrm{Cov}(X, Y) = E[(X - \mu\ )(Y - \mu\ )]$

One number computed from paired observations. Its sign is read in three ways; its size on its own is not the message.

SPLIT BY sign of $\mathrm{Cov}(X, Y)$

CASE 01 — $\mathrm{Cov} > 0$

Deviations move the same way

higher-than-average X tends to come with higher-than-average Y, and lower with lower

+ the mean product of deviations is positive

products $(X - \mu\ )(Y - \mu\ )$ are mostly > 0

X Y

EXAMPLE

height and weight across people

CASE 02 — $\mathrm{Cov} < 0$

Deviations move opposite ways

higher-than-average X tends to come with lower-than-average Y

− the mean product of deviations is negative

products $(X - \mu\ )(Y - \mu\ )$ are mostly < 0

X

EXAMPLE

outdoor temperature and heating use

CASE 03 — $\mathrm{Cov} = 0$

No consistent linear relationship

the deviations show no steady direction; this does **not** mean the variables are independent

0 the mean product of deviations is zero

the products cancel out

EXAMPLE

$Y = X$ with X symmetric about 0: Y is fixed by X, yet the covariance is 0